Automaton::DT_Automaton ::Initialization
+ std::string phase0
+ StateAction update0
+ double tau0
+ Initialization(std ::string phase0, StateAction update0, double tau0)
#initialization
Automaton::DT_Automaton
# State parameters
# std::set< std::string > inputs
# std::set< std::string > outputs
# State state
# std::map< std::string, Phase > phases
# std::vector< Transition > transitions
# std::string currentPhase
# double elapsed
+ virtual ~DT_Automaton ()=default
+ void addParameter(const std::string &name, std ::any value)
+ void addInput(const std::string &port)
+ void addOutput(const std::string &port)
+ void addStateVariable (const std::string &name, std::any initialValue)
+ void addPhase(const Phase &phase)
+ void addTransition (const Transition &transition)
+ void setInitialization (const Initialization &initialization)
+ void initialize()
+ double timeAdvance () const
+ bool externalTransition (const Event &x, double e)
+ std::vector< Event > internalTransition()
+ std::vector< Event > confluentTransition (const Event &x)
+ bool conditionHolds () const
+ const std::string & getCurrentPhase() const
+ const State & getState () const
+ const State & getParameters () const
+ static double infinity()
+ static std::string eventsToString(const std::vector< Event > &events)
- const Transition * find(TransitionKind kind, const Event *x) const
- std::vector< Event > fire(const Transition &t, const Event *x)
- void applyToState(const StateAction &f)
- void checkInput(const Event &x) const
Mu::ObserverData< T >
+ virtual ~ObserverData ()=default
+ virtual void update (const T &updatedData)=0
Mu::ObserverModel< T >
+ virtual ~ObserverModel ()=default
+ virtual void updateWithModel (Models::DT_Model< T > &m)=0
Data::DT_DataCharacterization
# std::string unit
# std::string systemStatus
# std::string signalType
# std::string dataFlowType
# std::string sensorPrecision
# std::string physicalLocation OfTheSensor
# std::string sensorCalibration Status
# std::string sensorLastMaintenanceDate
+ const std::string & getUnit() const
+ void setUnit(const std::string &value)
+ const std::string & getSystemStatus() const
+ void setSystemStatus (const std::string &value)
+ const std::string & getSignalType() const
+ void setSignalType (const std::string &value)
+ const std::string & getDataFlowType() const
+ void setDataFlowType (const std::string &value)
+ const std::string & getSensorPrecision () const
+ void setSensorPrecision (const std::string &value)
+ const std::string & getPhysicalLocationOfTheSensor () const
+ void setPhysicalLocation OfTheSensor(const std ::string &value)
+ const std::string & getSensorCalibrationStatus () const
+ void setSensorCalibration Status(const std::string &value)
+ const std::string & getSensorLastMaintenanceDate () const
+ void setSensorLastMaintenance Date(const std::string &value)
+ std::string toString () const
Data::DT_DataSegment< T >
- std::map< double, T > values
+ DT_DataSegment()=default
+ DT_DataSegment(double t, const T &d)
+ void put(double t, const T &d)
+ const T * get(double t) const
+ std::vector< double > getDomain() const
+ std::vector< T > getImage () const
+ const std::map< double, T > & getValues() const
+ bool isEmpty() const
+ std::size_t size() const
+ double infDom() const
+ double supDom() const
+ bool isRealTime(double tNow, double delta) const
+ bool isHistorical(double tNow, double delta) const
+ bool isMixed(double tNow, double delta) const
+ Kind kind(double tNow, double delta) const
+ DT_DataSegment< T > realTimePart(double tNow, double delta) const
+ DT_DataSegment< T > historicalPart(double tNow, double delta) const
+ DT_DataSegment< T > concat(const DT_DataSegment < T > &next) const
+ std::string toString () const
+ static std::string kindName(Kind kind)
Mu::ObserverDataManager< T >
+ virtual ~ObserverDataManager ()=default
+ virtual void registerObserver (std::shared_ptr< ObserverData < T > > observer)=0
+ virtual void removeObserver (const std::shared_ptr< ObserverData< T > > &observer)=0
+ virtual void notifyObservers()=0
Models::DT_Model< T >
# std::string modelName
- std::vector< std::weak _ptr< Mu::ObserverModel < T > > > servicesToBeUpdated
+ DT_Model(std::string modelName)
+ ~DT_Model() override =default
+ virtual void assimilate (const Data::DT_DataSegment < T > &omega)
+ virtual void executeModel()=0
+ virtual double evaluate (double t) const
+ void addServiceToBeUpdated (const std::shared_ptr< Mu::ObserverModel< T > > &service)
+ void removeServiceToBeUpdated (const std::shared_ptr< Mu ::ObserverModel< T > > &service)
+ void notifyServices()
+ std::size_t getNumberOfServices ToBeUpdated() const
+ const std::string & getModelName() const
+ void setModelName(const std::string &value)
+ virtual std::string toString() const
Services::DT_Service< T >
# std::string serviceName
# std::string serviceDescription
# std::shared_ptr< Models ::DT_Model< T > > model
# std::vector< T > consumedData
+ DT_Service(std::string serviceName, std::string serviceDescription)
+ ~DT_Service() override =default
+ void update(const T &data) override
+ void updateWithModel (Models::DT_Model< T > &m) override
+ Output execute(std ::shared_ptr< Models ::DT_Model< T > > m, const Data::DT_DataSegment < T > *omega)
+ virtual Output executeService()=0
+ const std::shared_ptr < Models::DT_Model< T > > & getModel() const
+ void setModel(std:: shared_ptr< Models:: DT_Model< T > > m)
+ const std::string & getServiceName() const
+ void setServiceName (const std::string &value)
+ const std::string & getServiceDescription () const
+ void setServiceDescription (const std::string &value)
+ const std::vector< T > & getConsumedData () const
+ virtual std::string toString() const
+ static std::string outputToString(const Output &output)
#characterization
-collectedData
Data::DT_Data< T >
# std::string source
# std::string dataType
- std::vector< std::shared _ptr< Mu::ObserverData < T > > > observers
- T currentData
- double currentTime
- DT_DataSegment< T > collectedData
- Clock clock
+ DT_Data(std::string source, std::string dataType)
+ ~DT_Data() override =default
+ void registerObserver (std::shared_ptr< Mu ::ObserverData< T > > observer) override
+ void removeObserver (const std::shared_ptr < Mu::ObserverData< T > > &observer) override
+ void notifyObservers () override
+ void updateData(const T &newData)
+ void updateData(double t, const T &newData)
+ const T & getCurrentData () const
+ double getCurrentTime () const
+ const DT_DataSegment < T > & getCollectedData () const
+ DT_DataSegment< T > getRealTimeData(double delta) const
+ DT_DataSegment< T > getHistoricalData(double delta) const
+ void setClock(Clock newClock)
+ bool validateData() const
+ const std::string & getSource() const
+ void setSource(const std::string &value)
+ const std::string & getDataType() const
+ void setDataType(const std::string &value)
+ DT_DataCharacterization & getCharacterization()
+ const DT_DataCharacterization & getCharacterization() const
+ void setCharacterization (const DT_DataCharacterization &value)
+ std::string toString () const
Models::DT_Model_Descriptive< T >
# std::string dataSource
# int dataPointCount
# std::string dataSummary
# double minimum
# double maximum
# double sum
+ DT_Model_Descriptive (std::string modelName, std::string dataSource)
+ void update(const T &updatedData) override
+ void executeModel() override
+ double getMean() const
+ const std::string & getDataSource() const
+ int getDataPointCount () const
+ const std::string & getDataSummary() const
+ std::string toString () const override
Services::DT_Service _Diagnosis< T >
# double normalMinimum
# double normalMaximum
# std::string diagnosisReport
+ DT_Service_Diagnosis (std::string serviceName, std::string serviceDescription, double normalMinimum, double normalMaximum)
+ DT_Service< T >::Output executeService() override
+ const std::string & getDiagnosisReport () const
std::enable_shared _from_this< DT_Mu< T > >
DT_DMS< T >
# std::string name
# std::vector< std::shared _ptr< Data::DT_Data< T > > > dataList
# std::vector< std::shared _ptr< Models::DT_Model < T > > > modelList
# std::vector< std::shared _ptr< Services::DT_Service < T > > > serviceList
# std::shared_ptr< Mu ::DT_Mu< T > > mu
# double delta
# std::shared_ptr< Data ::Clock > clock
- std::size_t dataIterator
- std::size_t modelIterator
- std::size_t serviceIterator
+ DT_DMS(double delta)
+ DT_DMS(std::string name, double delta)
+ DT_DMS(const DT_DMS &DT)=default
+ virtual ~DT_DMS()=default
+ double tNow() const
+ void setClock(Data ::Clock newClock)
+ double getDelta() const
+ const std::string & getName() const
+ void setName(const std::string &value)
+ void addData(const std::shared_ptr< Data ::DT_Data< T > > &d)
+ std::vector< Data:: DT_DataSegment< T > > getRealTimeData() const
+ std::vector< Data:: DT_DataSegment< T > > getHistoricalData () const
+ void addModel(const std::shared_ptr< Models ::DT_Model< T > > &m)
+ std::vector< std::shared _ptr< Models::DT_Model _Descriptive< T > > > getDescriptive Models() const
+ std::vector< std::shared _ptr< Models::DT_Model _Predictive< T > > > getPredictive Models() const
+ std::vector< std::shared _ptr< Models::DT_Model _Prescriptive< T > > > getPrescriptive Models() const
+ void addService(const std::shared_ptr< Services ::DT_Service< T > > &s, const std::shared_ptr< Models::DT _Model< T > > &m)
+ Services::DT_Service < T >::Output runService (const std::shared_ptr< Services::DT_Service< T > > &s, const Data::DT_DataSegment < T > *omega)
+ std::vector< std::shared _ptr< Services::DT_Service _Monitoring< T > > > getMonitoring Services() const
+ std::vector< std::shared _ptr< Services::DT_Service _Prediction< T > > > getPrediction Services() const
+ std::vector< std::shared _ptr< Services::DT_Service _Diagnosis< T > > > getDiagnosisServices() const
+ std::vector< std::shared _ptr< Services::DT_Service _Optimization< T > > > getOptimization Services() const
+ std::vector< std::shared _ptr< Services::DT_Service _Control< T > > > getControlServices() const
+ Mu::DT_Mu< T > & getMu()
+ void assimilate(const Data::DT_DataSegment < T > &omega)
+ bool checkServicesVersus Models(std::ostream &out) const
+ bool checkModelsVersusData (std::ostream &out, double epsilon=1e-3, double bigM =0.1) const
+ bool checkDataVersusServices (std::ostream &out) const
+ bool checkConsistency (std::ostream &out) const
+ std::string toGraphviz () const
and 21 more...
- Data::Clock sharedClock () const
- static std::vector < std::shared_ptr< U > > filter(const std ::vector< std::shared _ptr< V > > &list)
- static std::string block(const std::string &id, const std::string &title, const std::vector < std::string > &lines, const std::string &color)
- static std::string escape(const std::string &s)
Data::DT_Data_SineValues
- double anomalyTime
- double anomalyOffset
+ DT_Data_SineValues (const std::string &name)
+ double measure(double t)
+ void injectAnomaly (double time, double offset)
Models::DT_Model_Descr _CubicBSpline
# double step
# double leftEndpoint
# std::vector< double > controlPoints
+ DT_Model_Descr_CubicBSpline (double step, double leftEndpoint)
+ void update(const double &updatedData) override
+ double getSineModelValue (double x) const
+ double evaluate(double t) const override
+ double getLastTime () const
+ double getLeftEndpoint () const
+ double getStep() const
- void specifyAutomaton()
- double point(int i) const
Services::DT_Service _Diagnosis_Sine
# double threshold
# std::vector< double > anomalyTimes
+ DT_Service_Diagnosis _Sine(double threshold)
+ DT_Service_Diagnosis _Sine(const DT_Service _Diagnosis_Sine &)=delete
+ DT_Service_Diagnosis _Sine & operator=(const DT_Service_Diagnosis_Sine &)=delete
+ Output executeService () override
+ double performRandomCheck (int numberOfPointsToCheck, unsigned int seed, bool verbose) const
+ double getThreshold () const
+ const std::vector< double > & getAnomalyTimes () const
+ std::string anomalyTimes ToString() const
# double residualOf(double d) const
- void specifyAutomaton()
Mu::DT_Mu< T >
- double delta
- Data::Clock clock
- std::vector< std::shared _ptr< Models::DT_Model < T > > > models
- std::vector< std::shared _ptr< Data::DT_Data< T > > > sources
- std::vector< Data:: DT_DataSegment< T > > pending
- MixedPolicy mixedPolicy
+ DT_Mu(double delta, Data::Clock clock)
+ void addModel(const std::shared_ptr< Models ::DT_Model< T > > &m)
+ void observe(const std::shared_ptr< Data ::DT_Data< T > > &source)
+ Models::DT_Model< T > & mu(const Data:: DT_DataSegment< T > &omega, Models::DT_Model< T > &m)
+ void assimilate(const Data::DT_DataSegment < T > &omega)
+ void assimilateSynchronously (const Data::DT_DataSegment < T > &omega)
+ void assimilateAsynchronously (const Data::DT_DataSegment < T > &omega)
+ void flush()
+ double getDelta() const
+ std::size_t getPendingCount () const
+ MixedPolicy getMixedPolicy () const
+ void setMixedPolicy (MixedPolicy policy)
+ const std::vector< std::shared_ptr< Models ::DT_Model< T > > > & getModels () const
+ const std::vector< std::shared_ptr< Data ::DT_Data< T > > > & getSources () const
Visualization::DT_ChartPanel
- std::vector< double > xData
- std::vector< double > rawSineData
- std::vector< double > modelData
+ void updateChart(double index, double rawSine, double modelValue)
+ void writeCsv(const std::string &fileName) const
+ void writeSvg(const std::string &fileName) const
+ std::size_t size() const
- void polyline(std:: ofstream &out, const std::vector< double > &values, PX px, PY py, const char *color, const char *style) const
+sineData
+splineModel
+diagnosis
+assimilation
+chart
DT_DMS_SineCubicBP
+ static constexpr double STEP
+ static constexpr double DELTA
+ static constexpr double ANOMALY_THRESHOLD
- double now
+ DT_DMS_SineCubicBP (bool showChart)
+ void execute(int nbVal)
+ void writeGraphviz () const
+ void writeChart() const